ID No: ____________________ Name: ____ Sec No.: ______

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE,

PILANI

II SEMESTER 2025-26

A

CS/ECE/EEE/INSTR F241 Microprocessor Programming & Interfacing

24-04-2026 (Friday) QUIZ-6 (closed book)

Max time: 15 Min Max Marks: 08

Q1. An 8086-based interrupt-driven control system uses three 8259A PICs in cascaded mode: one Master PIC and two Slave PICs. The Master PIC is initialized with vector base address **40H**. Slave-1 is connected to IR2 of the Master and initialized with vector base address **70H**. Slave-2 is connected to IR5 of the Master and initialized with vector base address A0H. During operation, the following interrupt requests occur nearly simultaneously:

- IR4 of Master
- IR6 of Slave-1
- IR3 of Slave-2

Assume fully nested mode, edge-triggered interrupts, and no masking. Answer the following:

Question	Answer	
a. Which interrupt request will be serviced first? .[2M]	IR6 of Slave-1	
b If the ISR address for IR3 of Slave-2 is stored as IP = 2500H and CS = 9000H, write the four IVT memory locations and their contents for IR3 of slave 2. [2M]	Address (in hex)	Content (in hex)
	0028c	00
	0028d	25
	0028e	00
	0028f	90
c. What is the vector number of IR6 of Slave-1 (in hex)[2]	76h	

~~0000 0028c~~

0000 : 0028C

Q.2. An 8086-based system uses an 8254 timer. Counter 0 is programmed in Mode 2 (Rate Generator) with a clock input of 1 MHz. The counter is loaded with a count value of N = 1000.The output of Counter 0 is connected to a device that is sensitive to the duty cycle of the output waveform .Ignore the slight variation in duty cycle if any in the first period.

Question	Answer
a What is the duration of HIGH time of the output signal? (mention in microseconds) [1M]	999
b What is the duration of LOW time of the output signal? (mention in microseconds) [1M]	1

0000 : 028C 028C